MEMORANDUM

To: Operations Manager
From: Senior Mechanical Engineer
Date: April 1, 2026
Subject: Utility Water Analysis

EXECUTIVE SUMMARY

For the utility water transfer system, the flowrate of 400 L/min has been evaluated for pump power across the total discharge head. The findings are that there is a total discharge head of 127 m to overcome, which requires a power demand of 11.85 kW. The system is suitable for the increased flowrate, as the required power is certainly attainable with industry standard pump and motor. A good motor for this service would be the WEG W21Xdb 15 kW IEC Explosion-Proof Motor. It should be noted that the biggest bottleneck in the system is the 2-½" globe lift check valve and the long run of pipe along with its elevation. Further investigation into alternate check valves or possible pipe re-routing (shorter runs) could be explored if the client wanted to reduce overall head. A summary of the calculations to reach this conclusion follows below.

INPUTS

Input	Value	Source
Feedwater Internal Diameter (D_0)	D_0 = 600 mm	Feedwater Pipe Cross-Sectional Drawing
Relative Roughness of Feedwater Pipe (ϵ / d)	ϵ / d = 0.00036	A-23 of Appendix A (for 700 mm Cast Iron pipe)
Temperature of Water (T)	T = 20 °C	Prompt
Internal Diameter (D_2)	D_2 = 77.9 mm	B-16 of Appendix B (for 3" Sched. 40 pipe)
Internal Diameter (of the 2-½ segment of pipe) (D_1)	D_1 = 62.7 mm	B-16 of Appendix B (for 2-½" Sched. 40 pipe)
Water Density (ρ)	ρ = 998.2 kg/m^3	A-6 of Appendix A
Dynamic Viscosity (μ)	μ = 0.98 mPa-s	A-3 of Appendix A
Top of Elevation (Z_2)	Z_2 = 120 m	Piping System Diagram
Bottom of Elevation (Z_1)	Z_1 = 0 m	Piping System Diagram

FORMULAS

Name	Formula	Source
Darcy's formula	Q = 0.2087*d^2*sqrt((h_L*D) / (f*L))	Fluid mechanics
Hydraulic Radius	R_H = (cross-sectional flow area) / (wetted perimeter)	Fluid mechanics
Equivalent Diameter from Hydraulic Radius	D_equiv = 4*R_H	
Bernoulli's Theorem	Z_1 + (10^5*P_1) / (ρ_1*g_n) + (V_1^2 / (2*g_n)) = Z_2 + (10^5*P_2) / (ρ_2*g_n) + (V_2^2 / (2*g_n)) + h_L	Fluid mechanics
Pump Head (m)	H = (10^5 / ρ*g_n) * (P_1 - P_2) = (Z_2 - Z_1) + h_L	Re-writing Bernoulli's theorem, since ρ_1 = ρ_2 and V_1 = V_2
Head Loss (m)	h_L = 22.96*K*Q(in L/min)^2 / D(in mm)^4	Fluid mechanics
Reynolds Number	Re = 0.0053*(Q*ρ)/(R_H*μ)	Fluid mechanics
	Re = D(in mm)*V(in m/s)*ρ(in kg/m^3) / μ(in mPa-s)	
Velocity (m/s)	V = 21.22*Q(in L/min) / D(in mm)^2	Fluid mechanics
Power Demand (kW)	BHP = Q(in L/min)*H(in m)*ρ(in kg/m^3) / 6116*10^3*e_p	B-21 of Appendix B
Resistance Coefficient for Elbow	K_elbow = 30*f_T	A-29 of Appendix A (for a 90° elbow)

Name	Formula	Source
Resistance Coefficient for Gate Valve	K_gate = 8*f_T	A-27 of Appendix A (for a gate valve)
Resistance Coefficient for Straight Pipe	K_pipe = f*L/D	Fluid mechanics
Resistance Coefficient for Piping Exit	K_exit = 1.00	A-29 of Appendix A (for the exit)
Resistance Coefficient for Lift Check Valve w/ Reducer	K_check = (K_1+β*(0.5*(1-β^2)+(1-β^2)^2)) / β^4	A-27 of Appendix A
Diameter Ratio	β = D_1 / D_2	A-26 of Appendix A
K-factor Coefficient	K_1 = 600*f_T	A-27 of Appendix A

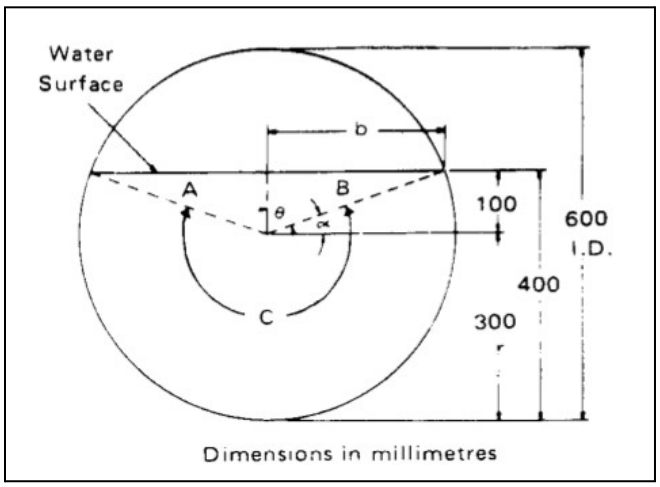
CALCULATIONS

1. Flow Rate Determination

Q = 0.2087*d^2*sqrt((h_L*D) / (f*L))

Since the pipe is flowing partially full, an equivalent diameter based upon hydraulic radius is substituted for D (D_equiv = 4*R_H) in the Darcy flow rate equation.
Q = 0.2087*D^2*sqrt((h_L*4*R_H) / (f*L)) = 0.4174*D^2*sqrt((h_L*R_H) / (f*L))

Now we need to determine what R_H (R_H = (cross-sectional flow area) / (wetted perimeter)) is:
Re = 0.0053*(Q*ρ)/(R_H*μ)



Depth of flowing water is 400 mm according to the Feedwater Pipe Cross-Sectional Drawing, but we need to calculate the flow area and wetted perimeter. Let's start with the flow area:
Cosθ = 100/300 = 0.333 θ = 70°32' α = 90° - θ = 90° - 70°32' = 19°28' = 19.47°

Area_C = (π*D_0^2 / 4)*(180+(2*α) / 360) = (π*(600 mm)^2 / 4)*(180+(2*19.47) / 360)
= 172,000 mm^2

b = sqrt(300^2 - 100^2) = 283 mm

Area_A = Area_B = (1/2)*(100*283) = 14,150 mm^2

Therefore, the cross-sectional flow area equals:
A+B+C = 2*14,150 + 172,000 = 200,300 mm^2 = 0.2003 m^2

D^2 = 4*a / π = 4*200,300 / π = 255,000 mm^2
(use this to substitute into the Darcy flow equation later)

h / L = Δh = 1 / 7.96*10^5 = 1.256*10^-6

Therefore, the wetted perimeter equals:
π*D_0*((180+(2*α)) / 360) = π*(600 mm)*(180+(2*19.47) / 360) = 1146 mm = 1.146 m

R_H = (cross-sectional flow area) / (wetted perimeter) = 0.2003 m^2 / 1.146 m = 0.175 m

Equivalent diameter based on hydraulic radius *(D_equiv)* determination:
D_equiv = 4*R_H*1000 = 4*(0.175 m)*1000 = **700 mm**

From A-23 of Appendix A (for 700 mm cast iron pipe):
ϵ / d = ϵ / D_equiv = 0.00036; f **= 0.0156**

Flow rate from the feedwater cast iron pipe to the pump *(Q)* determination:

Q = 0.4174*D^2*sqrt((h_L*R_H) / (f*L))

= 0.4174*(255,000 mm^2)*sqrt(((1.256*10^-6)*(0.175 m)) / (0.0156)) = 399.525

rounded to nearest whole number = **400 L/min**

Q (m^3/hr) = 399.525 L/min * (1m^3 / 1000 L) * (60 min / 1 hr) = 23.975

rounded to nearest whole number = **24 m^3/hr**

Q (bbl/d) = 399.525 L/min * (1 bbl / 159 L) * (60 min / 1 hr) * (24 hr / 1 d) = 3,618.34 bbl/d

rounded to nearest whole number = **3,618 bbl/d**

2. Loss Coefficient Determination

Friction factor *(f)* determination:

Velocity: V = 21.22*Q / D_2^2 = 21.22*400 / 77.9^2 = 1.4 m/s

Diameter Ratio: β = D_1 / D_2 = 62.7 / 77.9 = 0.80

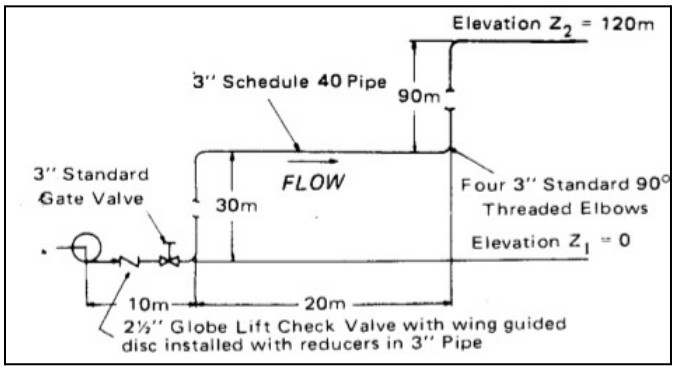
Reynolds Number: Re = D_2*V*ρ / μ = 77.9*1.4*998.2 / 0.98 = 1.1*10^5

Friction Factor, for pipe (f): f = **0.021** (based on Reynolds number, from A-25 of Appendix A)

Friction Factor, for valves / fittings (f_T): f**_T** = **0.018**

(from A-26 of Appendix A (for 2-½" or 3" size pipe))

e_p = 70% = 0.7



Resistance factors *(K)* determination:

K_elbow = 30*f_T = K = 30*0.018 = 0.540, but for 4 elbows, 4*0.540 = **2.16**

K_gate = 8*f_T = 8*0.018 = **0.144**

K_check = (K_1+β*(0.5*(1-β^2)+(1-β^2)^2)) / β^4 =

((600*0.018)+0.80*(0.5*(1-0.64)+(1-0.64)^2)) / 0.41

K_check = **27.0**

K_pipe = f*L/D = (0.021*(10 m+30 m+20 m+90 m))/(77.9 mm*(1 m/1000 mm)) = **40.4**

K_total = K_elbow + K_gate + K_check + K_pipe + K_exit = 2.16 + 0.144 + 27.0 + 40.4 + 1

= **70.7**

3. Total Dynamic Head Calculation

Using results from prior steps:

Total Discharge Head (TDH) Determination:

Pump Head Loss: h_L = 22.96*K*Q^2 / D^4 = (22.96*70.7*(400 L/min)^2) / (77.9 mm)^4 = 7.05 m

Total Discharge Head: H = TDH = (Z_2 - Z_1) + h_L = (120-0) + 7.05 = 127.05 m

Clear breakdown of each head contribution:

Bernoulli's Theorem: Z_1 + (10^5*P_1) / (ρ_1*g_n) + (V_1^2 / (2*g_n))

= Z_2 + (10^5*P_2) / (ρ_2*g_n) + (V_2^2 / (2*g_n)) + h_L

Since ρ_1 = ρ_2 and V_1 = V_2, re-write to establish pump head (H):

H = (10^5 / ρ*g_n) * (P_1 - P_2) = (Z_2 - Z_1) + h_L

Elevation head = Z_2 - Z_1 = 120 - 0 = 120 m

Head loss = h_L = 22.96*K*Q^2 / D_2^4 = (22.96*70.7*(400 L/min)^2) / (77.9 mm)^4 = 7.05 m

Total Discharge Head *(TDH)* determination:

TDH = H = (Z_2 - Z_1) + h_L = 120 + 7.05 = **127.05 m**

4. Pump Power Requirement

Power demand *(BHP)* determination:

BHP = Q*H*ρ / 6116*10^3*e_p

= (400 L/min)*(127.05 m)*(998.2 kg/m^3) / (6116*10^3*0.7) = **11.85 kW**

BHP (hp) = 11.85 kW * (1.341 hp / 1 kW) = **15.89 hp**

BHP (BTU/hr) = 11.85 kW * (3,412 BTU/hr / 1 kW) = **40,432.20 BTU/hr**

ENGINEERING RECOMMENDATION & MOTOR SELECTION

For the utility water transfer system, the flowrate calculated was 400 L/min. This rate has been evaluated for required pump power across the total discharge head. The total discharge head calculated to overcome is 127 m. This is to be expected and not anything out of the ordinary for a utility water transfer system in the oil & gas industry. It requires a power demand of 11.85 kW, quite a normal and readily available size for pumps & motors on an offshore oil & gas facility. Therefore, the system is suitable for this process, as the required power is certainly attainable with industry standard pump and motor.

It should be noted that the biggest bottleneck in the system is the 2-½" globe lift check valve and the long run of pipe along with its elevation. Further investigation into alternate check valves or possible pipe re-routing (shorter runs) could be explored if the client wanted to reduce overall head.

For BHP of 11.85 kW, the next standard motor size above that is 15 kW. For offshore oil & gas pumps, the motor should be at least explosion-proof, corrosion-resistant, TEFC. WEG is a major manufacturer of offshore oil & gas motors, they offer the WEG W21Xdb 15 kW IEC Explosion-Proof Motor, which will be sufficient for our application. Technical data on the specific motor can be found here: https://www.weg.net/catalog/weg/RS/en/Electric-Motors/IEC-Hazardous-Area-Motors/Flameproo f-Ex-db/Low-Voltage-Flameproof-Motors/W21Xdb---Flameproof-Ex-db/W21-Explosion-Proof-IE1 -15-kW-4P-160L-3Ph-380-415-660-440-460-V-50-Hz-IC411---TEFC---B5T/p/10083402.